YouTube Data Analysis – Detailed Analysis Explanation

Introduction & Objective :

The objective of this project is to analyze a global YouTube dataset to understand how subscriber growth varies across different channel categories, channel types, countries, and time periods. The analysis focuses on identifying patterns in subscriber distribution, recent growth trends, and the impact of channel creation timeline using exploratory data analysis techniques.

Importing Required Libraries:

In this step, essential Python libraries such as Pandas and NumPy were imported for data manipulation and numerical operations. Visualization libraries like Matplotlib and Seaborn were used to create meaningful graphical representations of the data. These libraries form the core toolkit for exploratory data analysis

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

Loading the Dataset:

```
df = pd.read_csv("Global YouTube Statistics.csv" , encoding = "latin1")
```

Basic inspection methods such as viewing the first few rows and checking column names were used to understand the available features. This helped identify important columns related to subscribers, channel category, channel type, country, upload count, and channel creation date.

Data Cleaning & Formatting :

Data cleaning steps were performed to handle missing values and ensure numerical columns were in the correct format. Display settings were adjusted to improve numerical readability, especially for large subscriber values. These steps helped prepare the dataset for accurate aggregation and analysis.

Subscriber Analysis by Channel Category :

The dataset was grouped by channel category to calculate the average and median subscriber count. This analysis helped identify which content categories generally attract more subscribers and highlighted differences between popular and niche categories

Subscriber Analysis by Channel Type :

Subscriber trends were further analyzed by grouping channels based on their channel type. Aggregations such as mean and median subscribers were calculated to compare how different channel formats perform in terms of audience size.

Country-wise Subscriber Distribution :
A country-wise analysis was conducted to understand how subscriber counts are distributed geographically. Stacked bar charts were used to visualize the contribution of different countries across content categories, revealing regional dominance in specific categories.

Upload Activity Analysis :
Total uploads were aggregated by channel type to evaluate content publishing behavior. This analysis revealed that higher upload counts do not always correspond to higher subscriber counts, indicating that content quality and audience engagement play an important role.

Time-Based Analysis Using Created Year :
Channels were grouped by creation year to analyze how subscriber growth varies over time. Average subscriber counts were calculated to observe trends in newer versus older channels and how platform maturity impacts growth

Monthly Trend Analysis :
A month-wise analysis was performed using the channel creation month to identify seasonal patterns in subscriber growth. Line plots were used to visualize trends, helping identify months with higher average subscriber gains.

Recent Subscriber Growth (Last 30 Days):
The analysis focused on subscribers gained in the last 30 days to understand recent growth behavior. Grouping by channel type and creation timeline provided insights into which types of channels are currently gaining subscribers more rapidly.

Conclusion:
This project demonstrates a complete exploratory data analysis workflow using real-world YouTube data. Through data cleaning, aggregation, visualization, and interpretation, meaningful insights were derived that showcase analytical thinking and practical use of Python for data analysis